# PRAKUL JAIN

AI/ML Engineer | Generative AI & LLM Applications

Jaipur, Rajasthan, India | +91 9256639008 | prakuljn3105@gmail.com

linkedin.com/in/prakuljn | github.com/Prakuljn | prakuljain-portfolio.netlify.app

## PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience designing and deploying Generative AI and LLM-powered systems for enterprise use cases. Skilled in building Retrieval-Augmented Generation (RAG) pipelines, multi-agent AI workflows, and LLM evaluation frameworks using LangChain and LangGraph, and in fine-tuning open-source LLMs with LoRA for domain-specific tasks. Experienced in end-to-end data engineering with Databricks, Delta Lake, Kafka, and SQL/PostgreSQL, and in shipping production-grade REST APIs with FastAPI on AWS with Docker and CI/CD pipelines. Currently building AI-powered applications and enterprise data pipelines at ATV Tech Solutions, with prior experience in data science and applied machine learning. Focused on translating complex AI models into reliable, scalable, business-ready systems.

## WORK EXPERIENCE

**AI/ML Engineer** | ATV Tech Solutions *12/2025 – Present | Jaipur, Rajasthan*

- Designed and developed Generative AI applications for enterprise use cases, building Retrieval-Augmented Generation (RAG) pipelines and multi-agent AI workflows with LangChain and LangGraph to automate complex, multi-step business processes.
- Built an LLM-as-a-Judge evaluation framework to systematically score and monitor LLM output quality across AI features, enabling data-driven prompt and pipeline improvements and cutting manual QA review effort by an estimated 30%.
- Fine-tuned open-source LLMs using LoRA for domain-specific tasks, improving response relevance by an estimated 15–20% over base-model outputs and reducing reliance on prompt-only approaches.
- Engineered enterprise data pipelines using Databricks and Delta Tables, integrating ETL processes with Kafka and Redis for reliable, low-latency data processing and downstream AI consumption.
- Developed and deployed backend services and REST APIs with FastAPI, implementing CRUD operations against PostgreSQL and SQL data stores to support AI-driven application logic.
- Containerized services with Docker and implemented CI/CD pipelines for automated build, test, and deployment on AWS, improving release consistency and reducing manual deployment effort.
- Applied AI evaluation pipelines and prompt engineering practices to continuously monitor and improve model output quality, reliability, and safety across production AI features.

**Data Science Intern** | Dotsquares *08/2025 – 11/2025 | Jaipur, Rajasthan*

- Supported data science workflows including data preprocessing, feature engineering, and exploratory analysis to inform business and product decisions.
- Assisted in building and evaluating machine learning models, contributing to model development and performance tuning under senior engineer guidance.
- Gained hands-on exposure to production AI workflows, including API-based integration of ML models into backend systems.

**AI/ML Engineer Trainee** | REGex Software Services *12/2024 – 08/2025 | Jaipur, Rajasthan*

- Built machine learning models using scikit-learn for business forecasting, applying hyperparameter tuning to improve predictive performance.
- Deployed ML models to production using AWS SageMaker and AWS Lambda, enabling scalable, serverless inference.
- Leveraged large language models, including Gemini and AWS Bedrock, with prompt engineering techniques to generate strategic business insights for stakeholders.

## KEY PROJECTS

**Enterprise Document Extraction & Evaluation Pipeline — Databricks**

- Problem: Manual extraction of structured data from unstructured enterprise proposal documents was slow and error-prone, and there was no scalable way to verify the quality of LLM-generated extractions.
- Built a Databricks-based ETL pipeline using Delta Tables and LLM-powered extraction to automatically parse and structure enterprise documents into usable data.

- Designed an LLM-as-a-Judge evaluation layer within the same pipeline to automatically score extraction accuracy and flag low-confidence outputs for review.
- Technologies: Databricks, Delta Lake, Python, SQL, LangChain, LLMs, prompt engineering, ETL pipelines.
- Impact: Reduced manual document-processing and review effort by an estimated 30–35% while improving consistency of extracted data for downstream systems.

**MindMorph — AI-Powered Learning Platform (Multi-Agent Workflow)**

- Problem: A single-agent LLM pipeline could not handle the multi-step, specialized reasoning required to power an adaptive learning experience (e.g., content generation, assessment, and feedback working together).
- Designed a multi-agent workflow using LangGraph to orchestrate specialized agents across the learning platform, with each agent handling a distinct stage of the pipeline.
- Implemented agent-to-agent communication using Kafka producer/consumer patterns, decoupling agents into independent services that publish and consume events asynchronously.
- Used Redis for low-latency shared state and caching between agents, improving coordination speed across the agentic workflow.
- Technologies: LangGraph, LangChain, Apache Kafka, Redis, Python, FastAPI.
- Impact: Enabled a scalable, event-driven multi-agent architecture for the learning platform, improving reliability and responsiveness of multi-step agent workflows compared to a single-agent baseline.

**RAG-Based AI Assistant — Dotsy-Bot**

- Developed a conversational AI chatbot integrating LangGraph and LangChain, powered by a Retrieval-Augmented Generation (RAG) pipeline for real-time contextual understanding.
- Built RESTful APIs using FastAPI with full CRUD operations and PostgreSQL for data management, and created and managed embeddings in a vector database for high-performance semantic retrieval.
- Technologies: LangChain, LangGraph, FastAPI, PostgreSQL, vector databases, RAG.
- Impact: Improved response accuracy and contextual relevance by an estimated 20% through prompt tuning and retrieval filtering.

**LoRA Fine-Tuned LLM for Domain-Specific Tasks**

- Problem: General-purpose LLM responses lacked the precision and tone required for domain-specific use cases.
- Fine-tuned an open-source LLM using LoRA (Low-Rank Adaptation) to specialize model behavior for a defined domain task while keeping compute and storage costs low.
- Technologies: LoRA, Hugging Face, Transformers, Python, prompt engineering.
- Impact: Improved output relevance and consistency by an estimated 15–20% versus the base model on domain-specific evaluation samples.

**AI-Powered ETL Pipeline**

- Problem: Raw enterprise data required significant manual cleaning and structuring before it could be used in AI applications.
- Engineered an ETL pipeline combining traditional data processing with LLM-powered extraction and transformation logic, using Databricks, Delta Tables, Kafka, and Redis for reliable, low-latency processing.
- Technologies: Databricks, Delta Lake, Kafka, Redis, SQL, Python, LLMs.
- Impact: Streamlined data readiness for downstream AI features, cutting manual data-wrangling effort by an estimated 25%.

**BeautySync Pro — Smart Salon Management**

- Developed a Flask-based salon management web application with a SQL backend for handling services and user data.
- Integrated an AI-powered chatbot using LangGraph, Serper, and Tavily with LLM APIs for real-time, web-enhanced responses.
- Technologies: Flask, SQL, LangGraph, Serper, Tavily, LLM APIs.

## TECHNICAL SKILLS

**Generative AI & LLMs:** LLMs, RAG, LangChain, LangGraph, Multi-Agent AI Workflows, LLM-as-a-Judge Evaluation, LoRA Fine-Tuning, Prompt Engineering, AI Evaluation Pipelines, Vector Databases

**Machine Learning & Deep Learning:** Supervised & Unsupervised Learning, Feature Engineering, Model Evaluation, Data Preprocessing, ANN, CNN, RNN, LSTM, TensorFlow, Hugging Face, Transformers

**Data Engineering:** Databricks, Delta Lake / Delta Tables, ETL Pipelines, Apache Kafka, Redis, SQL, PostgreSQL, MySQL

**Cloud & DevOps:** AWS (SageMaker, Lambda), Docker, CI/CD Pipelines, Git, Version Control

**Backend Development:** FastAPI, Flask, REST APIs, CRUD Operations, API Integrations

**Programming & Tools:** Python, SQL, GitHub, Jupyter Notebook, VS Code

## EDUCATION

**Bachelor of Computer Applications (BCA)** | Maharishi Dayanand Saraswati University
*07/2022 – 05/2025 | Ajmer, Rajasthan*

**Intermediate (CBSE)** | Makrana Public School
*04/2020 – 04/2022 | Makrana, Rajasthan*

## CERTIFICATIONS & ACHIEVEMENTS

- Google Cloud Vertex AI — Google Cloud
- HackCrux Hackathon — LNMIIT
- Python (Advanced) — HackerRank
- SQL (Advanced) — HackerRank

## LANGUAGES

- English — Full Professional Proficiency
- Hindi — Full Professional Proficiency